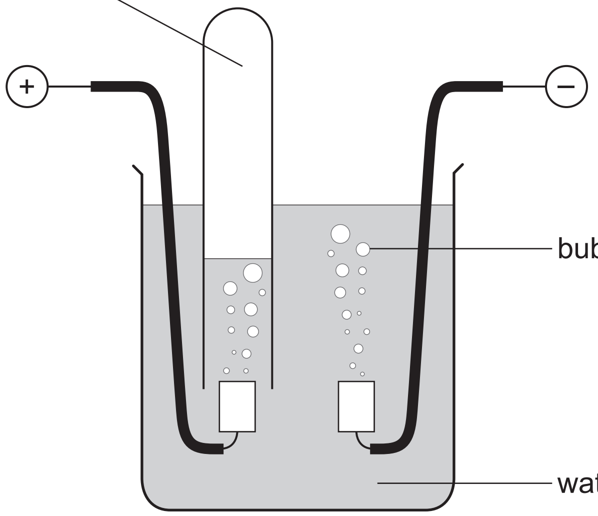
Examiner only

**10.** (a) The diagram shows the apparatus used by a student to investigate the electrolysis of water.




(i) The student collected a sample of the oxygen formed in a test tube.

Give the test the student would carry out to show the presence of oxygen in the test tube. Include the observation the student would expect. [1]

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(ii) Use the formula of water, $H_2O$.

Give the volume of hydrogen that would form in the same time as 10 $cm^3$ of oxygen. [1]

............................................ $cm^3$



 

Examiner only

(b) The table shows information about the electrolysis of three different electrolytes. The table is incomplete.

| Electrolyte | Ions present in the electrolyte | | Observations | |
|---|---|---|---|---|
| | Positive ion(s) | Negative ion(s) | At the negative (−) electrode | At the positive (+) electrode |
| molten lead(II) bromide | $Pb^{2+}$ | .................... | grey metal **A** formed | orange gas formed |
| aqueous copper(II) chloride | .................... and $H^+$ | $Cl^-$ and $OH^-$ | brown metal formed | green-yellow gas **B** formed |
| aqueous compound **C** | $Zn^{2+}$ and $H^+$ | $I^-$ and $OH^-$ | grey metal formed | brown solution formed |

(i) Complete the table by adding the **symbols** of the missing **ions**. [2]

(ii) Name substances **A**, **B** and **C**. [3]

Metal **A** ..............................................................................................

Gas **B** ..............................................................................................

Compound **C** ..............................................................................................

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